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J. Scott Holladay and Steven Soloway
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J. Scott Holladay¹ Steven Soloway²

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We examine the environmental and policy impacts of switching from oil-fired to natural gas-fired generation in New York City (NYC). We create an hourly panel of the fuel use of NYC's generators and use a semi-parametric approach to identify the fuel price spread that induces the switch from oil to gas. We find that NYC's pollution emissions decrease significantly after switching to natural gas. Around two-thirds of these emission reductions come from reduced emission intensity within plants, while the remaining third comes from less intense dispatch of oil fired generators. To illustrate the policy impact, we simulate the introduction of a real time pricing (RTP) program in NYC. The results suggest that the environmental benefits of the RTP decreased by nearly 30% due largely to fuel switching. While we focus on RTP, these results can be used to evaluate any energy policy that has a heterogeneous impact across time or the demand profile.

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¹ Assistant Professor, Department of Economics, University of Tennessee; jhollad3@utk.edu. Stokely Management Center, 916 Volunteer Blvd, Knoxville, TN 37919.

² Research Fellow, School of Law, New York University; steven.r.soloway@nyu.edu

1 Introduction

Electricity generation is one of the largest sources of air pollution in the United States. However, emissions have fallen consistently since 2009. Some of this reduction in pollution is because of regulations designed to reduce emissions and limit externalities, but a significant portion can be traced to unrelated price changes in other markets. Sustained low prices for natural gas have led many electricity generators to switch their fuel source from coal or oil to natural gas. Natural gas combustion produces lower levels of carbon dioxide, sulfur dioxide, and nitrogen oxides than the competing fuel sources. Therefore, the shift to natural gas has the ancillary benefit of reducing pollution emissions in the electricity sector.

This paper is an effort to determine the impact of the price changes in natural gas on the pollution emissions from electricity generators. We evaluate the environmental impacts of fuel switching at two margins: improved emissions intensity as oil fired-generators switching to natural gas (which we term the intensive margin) and less intense dispatch of pollution intensive power plants as lower cost natural gas generators are dispatched more intensely than oil-fired generators (the extensive margin). The flexibility of generators to burn multiple fuels has been widely studied.³ Pettersson *et al* (2013) analyze fuel switching in Western Europe and assess the impact of the carbon market on the incentives to switch fuels, but they do not look at the environmental impact of the switch. Maria *et al* (2014) finds that fuel choices did not change between the announcement and implementation of the US Environmental Protection Agency's (EPA) Acid Rain Program. Rather than focus on the impact of environmental regulation on fuel choice, we analyze the impact of fuel choice on environmental outcomes. For this analysis, we use the plausibly exogenous fall in natural gas prices caused by the widespread adoption of hydraulic fracturing and horizontal drilling for extraction.

A significant amount of literature has emerged that predicts the environmental impact of energy policy and highlights the importance of evaluating policy by using marginal emissions rather than average emission rates.

³ See Atkinson and Halvorsen (1976) for an early model of power plant fuel choice. Bopp and Costello (1990) provide a somewhat more recent review. Dahl and Ko (1998) estimates the impact of natural gas pipeline deregulation on fuel choice and compares their results to the previous literature. Soderh ldm (1999) provides a review of several empirical studies estimating price elasticities for power plant fuels at different points in time employing different methodologies and describes some of the difficulties for comparing estimated elasticities across the literature.

Cullen (2013) and Kaffine *et al* (2013) estimate the environmental impacts of expanding wind generation. Graf Zivin *et al* (2014) estimate marginal emissions across space and time and apply the results to evaluate the environmental impact of plug-in hybrid cars. This literature typically attempts to control for fuel prices using fixed effects and estimates a short-run marginal emission curve. In this paper, we estimate a medium-term emission rate with varying input prices but with a constant set of generators. This method highlights the importance of fuel costs in estimating marginal emission levels.

Fuel sources are imperfectly observed in various public data sets describing the electricity generating industry. To address this limitation, we build a unique data set that describes the electricity market in New York City (NYC) hour-by-hour from 2005-2010. We also take advantage of detailed generation and pollution emissions data reported to the EPA in conjunction with the SO₂ and NO_x markets implemented under the Clean Air Act. These data and the unique pollution signature of the various fuels allow us to use a series of semi-parametric regressions to identify the fuel price spread associated with generators switching to natural gas.

We then build an hourly panel of fuel type, electricity output, and pollution emissions for every generator in the five boroughs of NYC from 2005-2010. Over this period, we observe the oil-fired generators' transitions to natural gas and generation moving from oil to natural gas-fired generators. We estimate that when diesel prices exceed natural gas prices by more than around \$4 per million British thermal units (mmBTU), the fuel switching generators switch to natural gas. Using the panel nature of our data, we are able to estimate the pollution emissions averted by the shift to natural gas and natural gas-fired generators. These estimates can be used to evaluate the environmental impact of new and existing energy and environmental policies. To illustrate the use of these estimates, we simulate the impact of implementing real time pricing for electricity in NYC over the study period. The results suggest that the changing fuel price spreads reduce the environmental benefits of the real time pricing by around one-third. While we focus on real time pricing in our application, these results can be used to assess the environmental impact of any energy policy that has a heterogeneous impact across the demand profile such as plug-in hybrids, bulk electricity storage or energy efficiency building codes.

The paper proceeds as follows: in section 2, we provide background and describe the data; we describe the methodology for identifying the generators' fuel type in section 3; in section 4, we report the results of the semi-parametric estimation of the generation levels, fuel inputs, and pollution emissions. Section 5 presents the results of a simulation of a real time electricity pricing program in NYC. Section 6 describes the implications of the results and the final section concludes and provides some suggestions for future research.

2 Background and Data

New York City has a large and dynamic electricity market. For reliability reasons, the New York systems operator has instituted a local capacity requirement that mandates that within-city generators have the installed capacity to meet approximately 85% of NYC's forecasted peak load. The capacity requirement is enforced primarily through a monthly capacity auction.⁴ The auction winning units are required to be available to provide power each day of the contracted month. This requirement helps reduce the risk of catastrophic blackouts by lessening the city's reliance on imported electric power.

This process forces a significant amount of generation capacity to be located inside the city. This proximity exposes city residents to the pollution associated with electricity generation. The New York State regulations on acceptable emissions levels in the city have essentially banned coal-fired generation. This ban leaves a mix of oil- and natural gas-fired generation to meet the reliability requirement. Figure 1 describes the location, size, and pre-2005 fuel type of the generators in NYC. New York City is an ideal location to study the impacts of fuel switching on pollution because the city's changes in demand will affect the marginal generator that is also located in the city because of the reliability requirements. Any environmental or energy policy that affects demand differently across hours of the day or levels of generation will have an impact on the emissions of local generators.

Joskow (2013) describes how the twin technological innovations of hydraulic fracturing and horizontal drilling have led to huge decreases in the price of natural gas that have coincided with increased oil prices. Figure

⁴ For further reading on capacity market designs see Allcott (2012).

2 describes the evolution of oil and natural gas prices. Natural gas and oil prices both fell during the 2008 recession, but oil prices quickly recovered and began to slowly increase by early 2009. Natural gas prices continued to fall after the recession and have hovered near historic lows of less than \$2 per million btu since the spring of 2012. This combined effect is a significant relative price change in favor of natural gas.

In the long run, oil- and natural gas-fired capacity are near perfect substitutes, but once electric generators are constructed their ability to switch fuels is limited. For the most part, the boilers of electric generators cannot switch between coal and natural gas without cost once they are constructed (Söderholm (2001)). Switching from oil to natural gas as the primary generator fuel is a significant investment that requires one and half to two years of downtime (IEA (1988)) suggesting intensive margin impacts of fuel switching will require significant and stable changes in fuel prices.⁵ In the short run, fuel switching can occur at the extensive margin by bringing oil and natural gas generators online at different times in response to the changing marginal cost of production. No fuel type has enough generating capacity in NYC to meet the city's entire load so changing dispatch patterns cannot lead to complete fuel switching.⁶

An analysis of the fuel switching of flexible-fuel generators across both dimensions requires information on electricity generation, fuel prices, electricity markets, and the potential drivers of demand at high frequency. We construct a generator hour-level panel for every generator in NYC's five boroughs. We collect the data for our panel from the EPA's Clean Air Markets database. Their data is collected by the Continuous Emissions Monitoring Systems (CEMS) installed in every generator in the country with a generating capacity above 25 megawatts. The data is used to ensure compliance with the EPA's Acid Rain Program implemented under the Clean Air Act Amendments. The CEMS data reports hourly emission levels of SO_2 , NO_x , and CO_2 for each smokestack at every generator. The data also includes hourly measures of fuel inputs and electricity generation along with a host of

⁵ Fay *et al* (1986) reports that oil-to-gas conversions take less than a year and that converting a 50 megawatt generator could cost around \$1 million. More recent and detailed estimates of the cost of conversion are, to our knowledge, unavailable. Both studies are consistent with the anecdotal media reports on individual generator conversions: the costs and construction times are significant.

⁶ Other impediments to fuel switching include long-term fuel procurement contracts, the availability of pipeline capacity to provide fuel, and the potential triggering of a higher level of environmental regulation under the Clean Air Act's New Source Review due to major upgrades at existing facilities.

time-invariant information on the stack's location, ownership, and reported fuel sources. The fuel-source data lists a primary and secondary fuel source, but does not indicate which fuel is being used in which hour.

We collect fuel-price data from Thompson Reuters Datastream. The data include the daily closing prices for natural gas at Henry Hub and the prices of low sulfur 500 parts per million (ppm) diesel from NY Harbor.⁷ We combine this detailed generator and fuel-price data with a host of potential covariates. Weather is one of the largest drivers of variation in electricity demand. To control for that source of variation, we collect hourly weather data from NYC's Central Park Weather Observatory and the temperature, humidity, wind speed, and precipitation from the NOAA's National Climatic Data Center. New York State is part of a deregulated wholesale electricity market managed by the New York Independent System Operator (NYISO). We collect hourly data on the observed NYC electricity demand (load) from NYISO as well as the wholesale electricity prices. The full data set consists of an unbalanced panel of 472,278 generator-hour observations at 18 generators during the period of 2005 to 2010.⁸ Table 1 describes the variables and their summary statistics.

3 Identifying the fuel type

We focus on New York City as a whole and a two generators that use both natural gas- and oil-fueled generation throughout the study period.⁹ These fuel switching generators have the largest changes in marginal emissions rates, and policies that affect their fuel choice and production level are likely to have the largest environmental impacts. There are several publicly available data sets that can be used to identify what fuel a electricity generator is using. The (Energy Information Administration) EIA reports the total generation by fuel type monthly at the prime-mover/generator level.¹⁰ The EPA reports the primary and secondary fuel types each

⁷ Low sulfur diesel is used instead of the prices available for more highly polluting distillate fuel such as no. 6 Residual Oil. This choice is because the distillate fuels in the database contain sulfur that is far in excess of the 3,000ppm allowed by law in the dirtiest of residual oil.

⁸ Power plants that are not generating are not included in our dataset because we cannot identify a fuel type for those hours. Including those plants as zeros in the dataset rather than omitting those generator-hour observations does not materially affect the results.

⁹ One other power plant, Charles Poletti, responds to changing fuel prices primarily at the extensive margin, by reducing generation, rather than switching fuels.

¹⁰ Prime-mover is the technology that turns heat into electricity, for example, steam turbine, gas turbine, or combined cycle.

hour for each stack, but the data does not describe which fuel is in use at any given time. Neither of these data sets directly reports the fuel usage across load levels or times of day, so we are forced to estimate the fuel type from the emission rates in the EPA CEMS data. The data on which fuel type is in use each hour is crucial in order to link its emission level when estimating the environmental or energy policy outcomes.

Emission factors are published by the EPA in their AP-42 documentation.¹¹ These emission factors describe the expected pollution emission intensity per unit of fuel input for various types of fuel. By using the SO₂ emission rates in the EPA Clean Air Market data measured against the expected SO₂ combustion rates from the AP-42, the identification of the type of fuel a generator used during a given hour is possible. The primary aim of this exercise is to distinguish between oil and natural gas as a generation fuel, thus the SO₂ emission factors are sufficiently detailed to determine which type of oil was used as fuel.

Based on the SO₂ emission intensity calculated from the AP-42 data, determining when a generator chooses oil or natural gas is possible because there is a large difference in the SO₂ rates. Natural gas combusts at around 0.27 sulfur grams per millions of British Thermal Units (g/mmBTU) with very little variance. Distillate fuel oil number 4 combusts at about 460 sulfur g/mmBTU, and residual fuel combusts at 471 sulfur g/mmBTU. Most distillate fuels range anywhere from 450 to 462 sulfur g/mmBTU.

We define any SO₂ rate below 25 g/mmBTU, or 500 ppm sulfur, as clean middle distillate; any SO₂ rate of 100 g/mmBTU or less as middle to heavy distillate; and anything greater than 100 g/mmBTU as heavy distillate residual oil. In Table 2, we report average SO₂, NO_x and CO₂ emission rates for oil and natural gas fired generators. Figure 3 shows the emission intensity within the ranges of the oil- and natural gas-fired SO₂ emission rates. The distribution of natural gas-fired generation is extremely tight with a mean of 0.272, a fifth percentile of 0.272, and a ninety-fifth percentile of 0.273. The distribution of the oil-fired generation clearly shows differing emission intensities across oil types.

¹¹ The Compilation of Air Pollutant Emission Factors, or AP-42, "contains emissions factor and process information for more than 200 air pollution source categories." The latest edition of the AP-42 was published in 1995 and has been updated frequently.

We use the CEMS data to determine the fuel type for each stack at each generator in NYC for every hour from 2005 to 2010. We then aggregate the stack-level data by using the output measured in megawatts to create a summed distribution of the fuel use in each hour for each plant. The final data set is a generator-hour panel of generation, fuel type, and pollution emissions. The fuel-type choices for each plant are described in Table 3. The majority of the generators in NYC during the time period are exclusively natural gas-fired, but four plants rely on various types of oil. The plants that switch fuels can be identified by using the hourly CEMS data to break down how many megawatt-hours were produced with each fuel.¹²

Astoria Generating Station, Charles Polletti and Ravenswood Generating Station are the three large generators with significant oil generation over this time period. Those generators are the one with the ability to respond at either the intensive or extensive margin to changing fuel prices. Charles Polletti responds primarily by generating less as fuel prices increase. For that reason we we will focus on Astoria and Ravenswood in the empirical exercise below.

4 Empirical examination of fuel switching

4.1 Empirical approach

We estimate the extent of the environmental impact of fuel switching for NYC as a whole as well as for the two fuel switching generators identified in the previous section. We follow Holland and Mansur (2008) by using a semi-parametric model to estimate the marginal response of generators to changes in demand. We focus on the generation levels, fuel choices, and the marginal pollution levels across different levels of NYC loads to understand how the generator's fuel choice changes with different levels of demand. Analyzing the generator's production across demand levels is useful in evaluating the economic and environmental impacts of an energy policy.

We begin by analyzing the level of electricity generation across fuel price spreads and NYC demand levels.

¹² The cURL and Stata code to collect the data and identify the fuel type for each hour is available on the authors' websites.

The fuel price spread is defined as *diesel fuel* — *natural gas* price where both prices are measured in mmBTU.¹³

We conduct our analysis across price-spread quartiles, which are defined over the entire time series and summarized in Table 4.¹⁴ The lowest quartile ranges from -\$3.74 to \$4.39 and is observed primarily in 2005, but the spread dips to that level as late as 2009. The second quartile ranges from a spread of \$4.40 to \$7.54, the third quartile goes up to \$9.68, and the fourth is any spread above \$9.68. Table 5 summarizes the distribution of spread quartiles across time. From 2005 through the end of the data set in 2010, the spread increases consistently but the seasonal demand shifts and market shocks generate variations in the spread quartiles over time.

To control for varying energy market conditions, we define NYC electricity demand deciles using data from the NYISO.¹⁵ Not surprisingly, low deciles of load are observed primarily in the overnight hours, and higher load deciles occur primarily during peak demand hours (hours 7 a.m.- 11 p.m.).

We estimate the equation below for all generators in NYC and for Astoria and Ravenswood separately:

$$Gen_{f,h} = +\beta_{q,d} SpreadQuartile_h * NYCLoadDecile_h + \omega W_h + \delta_{f,h} + \eta + \varepsilon, \quad (1)$$

where f indexes facilities, and h denotes hours. The dependent variable, $Gen_{f,h}$, is generation at electricity generating facility f in hour h. $SpreadQuartile_h$ is an indicator variable for the *diesel fuel* — *natural gas* price-spread quartile, and $NYCLoadDecile_h$ is a set of dummy variables for NYC load deciles. The $\beta_{q,d}$ is a vector of 40 coefficients that describes each price-spread quartile(q)-NYC load decile (d) pair. W_h is a vector of weather controls that comprises the temperature, temperature squared, humidity, humidity squared, the interaction of the temperature and the humidity, and the temperature interacted with the amount of sunlight during the day measured in minutes.¹⁶ These variables control for the observable weather shocks that might

¹³ We have also explored using Algonquin natural gas prices and WTI oil prices to define spread quartiles based off New York City prices and off national prices to deal with potential endogeneity. The results are not sensitive to the definition of spread quartiles.

¹⁴ We include the fuel price spreads defined as quartiles rather than directly including the spreads in the regression to allow for the highly nonlinear impacts of the changing spreads. These quartiles allow us to separate the portion of the distribution where the largest impact of changing fuel prices is observed.

¹⁵ We conducted a similar analysis across hours of the day, rather than demand levels. This approach might be more suitable for certain types of energy policies. The results are qualitatively similar to those estimated across the NYC load levels.

¹⁶ The amount of daily sunlight is not a weather variable but serves as a continuous proxy for the time of year and allows a 90-degree day in August to have a different impact on generation than a 90-degree day in October.

affect electricity demand and the generator's dispatch choices within an NYC load decile.¹⁷ The δ is a set of month fixed effects, and the η is a set of year fixed effects that flexibly control for seasonal and long-run unobservables in electricity consumption and dispatch patterns respectively and ε is the error term.¹⁸ The β vector contains the coefficients of interest. They describe the marginal change in natural gas generation, from a 1 megawatt-hour increase in generation at a generator.

After analyzing the dispatch patterns, we move on to evaluating the fuel choices as the fuel price spread and demand change. To estimate these changes we modify equation 1 slightly and estimate:

$$Gas_{f,h} = +\beta_{q,d} * Gen_{f,h} * SpreadQuartile_h * NYCLoadDecile_h + \omega W_h + \delta + \eta + \varepsilon, \quad (2)$$

where the dependent variable is the level of generation, measured in megawatt-hours, from natural gas at the electricity generating facility. We use the methodology described above to define the fuel choice at the stack level and aggregate up to the generator level for each hour. The

$Gen_{f,h} * SpreadQuartile_h * NYCLoadDecile_h$ is a triple interaction of the electricity generating facility's generation level, fuel price spread, and NYC load. The $\beta_{q,d}$ coefficients now report the fraction of each megawatt-hour of additional generation from a generator fueled by natural gas across levels of fuel price spreads and NYC demand.

Next, we estimate the environmental impact of the fuel choice observed in the previous estimation. We reestimate equation 2 by using the observed SO_2 emissions measured in pounds of emissions. The $\beta_{q,d}$ coefficients estimate the marginal SO_2 emissions from an additional megawatt-hour of generation at that generator. This vector of coefficients can be used to assess the environmental impact of an energy policy. Each β provides the change in emissions associated with a marginal change in generation at that demand level and fuel price spread. Policies that affect demand at different levels of consumption can be assessed independently

¹⁷ Weather controls also allow for the impact of temperature on the efficiency of fossil fuel electricity generators as in Cullen (2013) and other papers in this literature.

¹⁸ The results are robust to including day of week and hour of day fixed effects as well.

from these results. This approach is consistent with the emerging literature on the importance of evaluating an energy policy based on marginal emissions and controlling for variation in marginal emissions across space and time.

4.2 Empirical results

4.2.1 Generation

Figure 4 shows the results of estimating equation 1 across all of the NYC generators (top panel), the Astoria Generating Station (middle panel), and the Ravenswood Generating Station (bottom panel).¹⁹ Each graph displays the 40 estimated β coefficients with NYC load deciles across the horizontal axis and spread quartiles connected by lines.²⁰ The vertical axis is the level of generation measured in megawatt-hours. The vertical bars represent the 95 % confidence interval for the coefficient estimate. For the regressions across all of the generators in NYC, the standard errors are clustered at the generator level. This clustering allows for generator specific serial correlation in the error term. For generator specific regressions we report Newey-West standard errors with 24 lags.²¹

The top panel shows little change in the average generation level within a NYC load decile across the four quartiles of fuel price spreads. The average level of dispatch across the city seems to be slightly higher in load deciles 6-9, although these differences are not statistically significant. The average level of generation weakly increases with NYC load levels, although the controls for the weather appear sufficient to soak up most of this variation. The average level of generation across the entire data set is 260.8 megawatt-hours per generator. The results suggest that the pattern of imports and exports of electricity do not vary significantly over the quartiles of fuel price spreads.

¹⁹ We focus on the two generators identified as switching fuels during the study period to better understand the intensive margin of fuel switching. Results for all generators in New York City are available by request.

²⁰ Decile 1, quartile 1 is the omitted category. It is set to zero in these graphs to aid in the interpretation of the other coefficients.

²¹ The cluster robust standard errors are consistent in the number of clusters. With 18 generators the robust standard errors may understate the true standard errors. We prefer to report cluster-robust standard errors where possible as they place (slightly) less structure on the nature of correlation in the error term. The results with Newey-West standard errors are, like the cluster-robust results, highly significant.

The next two panels (for Astoria and Ravenswood respectively) show the changes in dispatch at those generators. Using a series of t-tests, we find that Astoria generates significantly less electricity in the NYC load deciles of 5-10 in fuel price spread quartile 1 as compared to quartile 2. For fuel price spread quartiles 3 and 4, the difference is similar and statistically significant at the higher load deciles. The generation levels are higher in quartile 4 than quartiles 2 and 3, although the magnitude of that difference is relatively small. The results suggest that Astoria dispatches much more intensely when the fuel price spread is relatively small and that difference increases with the load levels. As demand increases, the relatively high priced generation from Astoria is brought online.

The bottom panel shows that the dispatch level for Ravenswood is also similar across fuel price spreads at low levels of demand. From load deciles 5-10, the generation levels are significantly higher during the period of relatively low price spreads. Spread quartiles 3 and 4 are statistically indistinguishable from each other with the exception of load decile 10.

4.2.2 Fuel choice

Figure 5 describes the results of estimating equation 2 across all of the NYC generators (top panel), Astoria (middle panel), and Ravenswood (bottom panel). The coefficients in this regression represent the marginal change in natural gas generation, measured in megawatt-hours, from a 1 megawatt-hour increase in generation at a generator. Quartile 1 of fuel price spreads in which natural gas is relatively more expensive has by far the lowest megawatt-hour increase in generation. In each price-spread quartile, the natural gas used to meet load falls as demand increases, but the decrease is much larger in quartiles 2-4.

For the city as a whole (top panel), the results of an F-test of the hypothesis that $\beta_{1,d} = \beta_{2,d}$ ²² for each load decile suggest that natural gas-fired generation is significantly lower in every decile but the tenth. In the highest load deciles, all of the available generation in the city is likely to be on and little or no difference is expected.

²² We conduct ten hypotheses tests, one for each NYC load decile, that compare the coefficient from the first price-spread quartile to the second.

Astoria (in the second panel) also produces less electricity from natural gas in price-spread quartile 1 while the other quartiles are (largely) statistically indistinguishable from each other. In load decile 1 in price-spread quartile 1, just over 40 % of a marginal increase in generation is served by natural gas. In the same load decile in price-spread quartile 2, the marginal increase rises to nearly 60 % . As with NYC as a whole, the fuel usage converges in demand decile 10 where the plant is operating at full capacity. Ravenswood (in the bottom panel) follows a similar pattern. Results of an F-test of the hypothesis that $\beta_{1,d} = \beta_{2,d}$ suggest that Ravenswood produces statistically much less electricity from natural gas in every load decile except the tenth.

The Astoria and Ravenswood generators did not add or close any generating units during the study period.²³ Using the regression results from the bottom two panels of Figure 5, we can discern the dispatch decisions consistent with the observed marginal generation profile. In price-spread quartile 1, both generators rely on less natural gas-fired generation as demand rises. In quartiles 2-4, they rely on increasing levels of natural gas-fired generation and bring more oil-fired generation online only at the higher demand levels. This pattern is particularly pronounced for Ravenswood that brings on increasing levels of natural gas until load decile 8 when the it is forced to greatly increase oil-fired generation.

4.2.3 Pollution

We analyze the impact of increases in generation on pollution emissions across load levels and relative price spreads. Figure 6 is a summary of the marginal SO₂ emissions associated with a marginal increase in generation across price-spread quartiles and NYC load deciles. The top panel, describing results for all generators in NYC, shows that price-spread quartile 1 has the highest emission level. Marginal emissions are statistically much larger at all of the load levels. A 1MWh increase in generation is associated with a 1.3 pound increase in emissions in load decile 1 that increases to 1.7 pounds in load decile 10. The point estimates are trending up throughout

²³ Each generator did change ownership during the study period. In the appendix, we present a robustness check that suggests this change had no impact on the way the units were dispatched.

price-spread quartile 1, but they are not estimated precisely enough to confirm that the trend is statistically significant. The marginal emission estimates are very similar across the other three price-spread quartiles. Once again, there is a weak upward trend in the emission rate as generation increases.

The second panel of Figure 6 shows the marginal emissions at Astoria. In the lowest price-spread quartile, Astoria has a slightly lower SO₂ emission rate than NYC as a whole. The emissions increase through the 7th quartile after which they drop slightly as more natural gas comes online. During the other three price-spread quartiles, the emission rates weakly increase as more oil-fired generation is brought online in the highest demand deciles. The bottom panel characterizes the marginal emissions at Ravenswood. Once again, the emission rates are significantly higher in the lowest price-spread decile. The emission rates are comparable to NYC as a whole and increase with demand in that quartile. Consistent with the gas-generation profile in Figure 5, the emission rates increase sharply in the highest two demand deciles as large quantities of oil-fired generation are brought online.

We turn now to estimating the SO₂ emission reductions at the extensive and intensive margins. We define the extensive margin as market driven fuel switching due to changes in dispatch intensity of oil and gas fired generation. These “fuel switches” occur as the relatively high-priced oil generation at the generators is displaced by natural gas generation. We define the intensive margin as within plant changes in emissions intensity from fuel switching. The intensive margin changes could be driven by changes in the infrastructure of a plant from oil to gas, changing fuels at flexible fuel units or changing dispatch intensity of boilers within a plant.²⁴ In this paper we abstract from the mechanism and refer to all within plant changes in fuel use and emissions intensity as the intensive margin

In Table 6, we summarize the total emissions and generation between quartile 1 and the average of quartiles 2-4 for each generator in NYC. The aggregate emissions fall by 25.8 million pounds from quartile 1 to quartiles 2-4. Eighty-five% of that net decrease occurs at Astoria and Ravenswood: 37% and 48% respectively. Six

²⁴ In the appendix we provide evidence that the emission reductions are not primarily driven by changing dispatch of units at a given plant.

plants increase their aggregate emissions that account for a total of 15,176 pounds of the increase.²⁵

The total generation in NYC falls by 1.3 million megawatt-hours (around 4 %) between the first and second to fourth quartiles. The reductions in generation at Astoria and Ravenswood are 1.9 million and 2.7 million megawatt-hours respectively. We evaluate the extensive margin by calculating the change in generation between quartile 1 and quartiles 2-4 multiplied by the emission rate in quartile 1. We calculate the intensive margin by calculating the change in the emission rates between quartile 1 and quartiles 2-4 and multiplying by the generation level in quartiles 2-4. These two margins sum to the total change in generation. Both fuel switching plants respond primarily at the intensive margin with nearly two-thirds of the reductions in emissions. Changing fuel prices drive fuel switching generators to lower emission intensities that lead to large reductions in the total emissions from electricity generation. The adjustments at the intensive margin for these two plants account for 55% of the total emission reductions across NYC observed between quartile 1 and quartiles 2-4.

4.3 Robustness checks

We conduct a number of robustness checks in the appendix to ensure that the observed generation and pollution are driven by relative fuel prices instead of some other unobserved factor that varies with fuel prices. These checks comprise controlling for changes in generator ownership, analyzing the impact of fuel switching on the emissions of other pollutants, using intra-month variation to identify impacts, results for other generators in NYC, and using price-spread deciles instead of quartiles. In this section, we focus on the most likely confound of the analysis: pollution permit prices. The fuel prices analyzed so far represent the largest variable cost in electricity production, but both SO₂ and NO_x emissions are regulated under cap and trade systems. The emission intensity of oil-fired generation is higher than natural gas meaning that the fuel price spread understates the full variable cost spread. If changes in emission prices vary systematically in a way that makes natural gas more attractive

²⁵ All of the results in this section refer to a net emission decrease, not the gross.

relative to oil, then this variation might be the source of the observed fuel switching.

Table 7 shows the pollution permit prices for SO₂ and NO_x.²⁶ The price of permits falls as the price spread increases because as oil (and coal) become less competitive they require fewer permits. The producers who sell their permits on the market have caused an extended period of extremely low permit prices. The regulatory uncertainty around the SO₂ trading program has also helped keep prices low.²⁷ This price drop has actually served to make oil *more* attractive relative to natural gas. Absent the changes in the permit price the shift to oil might have happened at even lower price spreads than those observed.

To confirm this result, we reestimate equation 2 with SO₂ emissions as the dependent variable and added controls for SO₂ and NO_x permit prices. The results, reported in Figure 7 suggest that controlling for pollution permit prices has little effect on the estimated emission intensity. The point estimates of the marginal emission intensity are slightly higher and more precisely estimated after controlling for the permit price. None of the differences is statistically significant, and the change in magnitudes is quite small. These generators are also subject to the Regional Greenhouse Gas Initiative (RGGI) that came into force in January of 2009. We reestimate equation 2 on the subset of the data before 2009. The results are consistent with the full data set, which is not surprising. The RGGI prices (in the range of \$2 to \$4 per ton) are swamped by the drop in relative spread prices.

5 Application to Real Time Pricing

The variation in marginal emissions over the load profile and fuel costs makes estimating the environmental impact of the proposed electricity market regulations *ex ante* extremely difficult. To illustrate this point, we evaluate the environmental impact of real time pricing on pollution emissions in NYC across different fuel prices.

Real time pricing seeks to incentivize electricity consumers to conserve at times of high demand by forcing

²⁶ We purchase the data on pollution permit prices from BCG Partners that is a dealer and broker in pollution permits.

²⁷ See Schmalensee and Stavins (2013) for a description of the functioning of these markets and an explanation for the price volatility.

households to face variable prices. The variable prices are related to the wholesale electricity costs at the time of consumption instead of the constant price throughout the monthly billing cycle that is based on long-run average costs of procuring electricity. A great deal of literature finds that real time pricing can have significant impacts on energy consumption patterns (see Borenstein (2005) for example). Real time pricing is typically implemented to equate electricity prices and marginal costs, but it can have significant environmental impacts.

Holland and Mansur (2008) finds that reducing the variance in electricity demand can reduce pollution emissions in regions that use oil-fired generation to meet peak demand. In related work, Holland and Mansur (2006) finds that real time pricing can increase aggregate load, but reduces profits of oil fired generators. The environmental impacts are mixed in their simulations with SO₂ and NO_x emissions increasing, but CO₂ emissions decreasing. Allcott (2011) reports the impact of real-time electricity pricing on household demand from a large scale field experiment in Chicago. The results suggest that real time pricing reduces consumption during high price hours without significant increases in demand during off-peak hours. Allcott's calculations suggest that real-time electricity pricing in the Chicago region could be associated with reductions in CO₂ emissions on the order of 4 % . To date the real-time pricing literature has not considered the impact of fuel prices on emissions.

To simulate the impact of real time pricing in NYC, we need estimates of the demand response to variable intraday prices. Allcott (2011) calculates the demand impact of real time pricing for hours of 6 a.m. to 9 p.m. for the summer and the rest of the year separately.²⁸ We follow Allcott and use hourly real-time wholesale electricity prices from NYISO for NYC to estimate the demand impact of real time pricing for each hour in our sample period. To compile this estimate we calculate:

$$DemandShift_{h,s,q} = \nu_s^A * \overline{price}_{h,s,q} + \nu_{h,s}^D (price_{h,s,q} - \overline{price}_{h,s,q}) + \nu_s^T, \quad (3)$$

where $\overline{price}_{h,s,q}$ is the average hourly wholesale electricity price in the NYC zone during hour h in season s and price-spread quartile q. This term captures the consumers' response to the expected electricity price.

²⁸ In Allcott's experiment there is too little variation in electricity prices over the nighttime hours to generate any change in electricity consumption. The experiment also includes a high-price alerting system that we omit for simplicity.

The $(price_{h,s,q} - \overline{price}_{h,s,q})$ term captures the deviations between the average price in hour h and season s and the actual observed price. The final T_s term captures the impact of being on real-time pricing plans generally.²⁹

Each of the ν parameters is taken from Allcott's estimates.³⁰ The resulting estimates are the impact of real time pricing per household measured in watts. To scale these estimates for NYC, we multiply by the three million NYC households as reported by the Census.³¹ Finally, we divide the resulting changes in electricity demand by one million to get estimates in megawatt-hours.

This procedure generates an estimated load reduction for each hour as if NYC had implemented a real-time pricing scheme. The results, averaged by the price-spread quartile and season are reported in Table 8. There is significant intraday variation in the demand changes, and those changes vary across price-spread quartiles. Summing the estimated emission reductions suggests that real time pricing should reduce the aggregate demand over the course of a year by 4.2 million megawatt-hours during price-spread quartile 1 and 3.9 million megawatt-hours during price-spread quartile 4.³² The lower electricity prices due to relatively inexpensive natural gas reduce the impact of the real time pricing on the aggregate demand by around 6%.

Because the real time pricing policy has different demand impacts in different hours of the day, we estimate the marginal emissions across each hour of the day for each price-spread quartile. We reestimate equation 1 with hour-of-day fixed effects instead of load deciles to facilitate this policy analysis. Specifically, we estimate:

$$Emissions_{p,h,f,s} = +\beta_{q,h,s} SpreadQuartile_q * Hour_h + \omega W_h + \delta + \eta + \varepsilon, \quad (4)$$

where $Emissions_{p,h,f,s}$ are the emissions of pollutant p (we focus on CO_2 , but the appendix reports results for SO_2 and NO_x as well) from electricity generating facility f in hour h . The $Hour_h$ is a set of hour-of-day

²⁹ By calculating the average price within a price-spread quartile, we are implicitly assuming that consumption patterns will respond to variations in the electricity prices driven by changes in relative to fuel prices.

³⁰ Allcott's estimates are in cents per kilowatt hour. We multiply by ten to transform those coefficient estimates into dollars per megawatt-hour to be consistent with the EPA's CEMS data and the NYISO's pricing data.

³¹ The American Community Survey's five-year estimates for 2007-2011 report 3,049,978 households in NYC and 3,371,062 housing units.

³² The sum of demand reductions requires multiplying the hourly estimates for summer by 92 days and the non-summer estimates by 273 days and adding the resulting products.

fixed effects. For consistency with the demand elasticity estimates, we estimate the equation separately for summer and non-summer months. The parameters of interest are the interaction effects of the twenty four hours and four price-spread quartiles that represent the marginal emissions from generation changes in a given hour and price-spread quartile. Those estimates are presented in Figure 8. The results are consistent with the previously discussed estimates of the marginal emissions across the demand profile. During spread quartile 1 (with expensive natural gas), the marginal emissions are at their highest overnight when demand is the lowest and decreases through the day as demand increases and more expensive natural gas is brought online. In the other price-spread quartiles, this pattern is reversed as natural gas serves a higher proportion of the base load, and relatively more expensive oil is brought on during higher demand hours.

To estimate the environmental impact of the counterfactual demand changes, we take each hour's estimated demand shift under real time pricing and multiply that by the marginal emissions of generation during that hour.³³ The product is the estimated environmental impact of real time pricing for an hour in a season in a price-spread quartile. Table 9 summarizes the results for CO₂ emissions. There is significant variation in all three dimensions: intraday, across seasons, and across fuel price spreads. Consistent with the literature, there are small environmental benefits to real time pricing. The benefits are largest over peak demand hours and during the summer. Moving outside of spread quartile 1 (the most expensive natural gas regime) reduces the benefits by around a third. The movements across the other spread quartiles lead to much smaller changes in environmental benefits.

The results presented here represent a short-run environmental response to real time pricing. In the long-run, consumers are likely to respond to real time pricing by adopting new technologies that allow them to better control electricity consumption throughout the day leading to changes in the demand elasticity parameters. Real time pricing could also change the way electricity generators are dispatched and/or the types of new generators

³³ This approach assumes that the changes in demand due to the real time pricing are in fact marginal changes. The hourly reductions are on the order of 5 to 10% of the total demand, which are the same magnitude as the changes in demand between hours during the sample period.

that are constructed in order to alter the prices and the demand-side response.

6 Discussion

The results above suggest that the real time pricing in electricity markets could have small, but significant, environmental benefits. The benefits of real time pricing are reduced by changes in relative fuel prices. As oil-fired electricity generators switch to natural gas and market forces change the dispatch of natural gas generators, electricity generation in NYC has become significantly cleaner. This effect is offset somewhat by oil-fired generation's shift to peak hours that has made these high demand hours relatively dirtier compared to low demand periods.

A full welfare analysis of real time pricing is beyond the scope of this paper, but to assess the economic significance of the results we can value the estimated CO₂ emissions offset by the real time pricing in NYC. To monetize the damages avoided, we use the EPA's estimated marginal damages of a ton of carbon, known as the social cost of carbon (SCC). Recent estimates using a 3 % discount rate suggest that the SCC will be \$39 in 2015. We use this value to estimate the benefits of emission reductions. Our estimates suggest that real time pricing could reduce CO₂ emissions by -2,768,901 tons in price-spread quartile 1 and -1,964,762 in price-spread quartile 4.³⁴ With a SCC value of \$39 per ton, this amount equates to annual benefits of \$107,987,145 and \$76,625,727 respectively. These benefits, while large in magnitude, are small on a per household basis with benefits moving from \$36.00 to \$25.54. The estimated costs of installing real-time pricing hardware vary from \$200 to \$1,000 per household that suggests the nonmarket benefits of CO₂ emission reductions alone are not sufficient to justify introducing real time pricing.³⁵ Allcott (2011) suggests the environmental benefits of real time pricing are small relative to the private benefits from reduced electricity consumption. A full cost-benefit analysis would require

³⁴ Results for price-spread quartiles 2 and 3 are -1,974,583 and -2,007,748 respectively.

³⁵ Including the reductions in SO₂ and NO_x emissions (discussed in the appendix) increases the nonmarket benefits of installing real time pricing significantly. The benefits from avoiding the installation of additional peak generating and transmission capacity could be very large as well. A full cost-benefit analysis would require estimating these other non-pecuniary benefits along with the external benefits.

estimating other non-pecuniary benefits along with the external benefits.

Of more relevance to this study is the variation in the estimated benefits across the price-spread quartiles and the hours of the day. Moving from price-spread quartile 1 to 4 reduces the CO₂ emission reduction benefits of real time pricing by 29%.³⁶ This reduction in benefits is driven largely by reductions in the marginal emission rates across quartiles. The demand reduction under real time pricing is attenuated by only 6%, which moves from price-spread quartile 1 to 4. The generation-weighted average marginal emission reduction is 26%. We know, from the analysis in section 4.2.3 that around two-thirds of these reductions in marginal emission rates are driven by reduced emission intensity, and the remaining third are driven by less intense dispatch at the pollution intensive generators. The hourly benefits can vary by as much as a third across the days within a price-spread quartile. Early morning hours produce the smallest benefits while hours 5 p.m. and 6 p.m. produce the largest across all of the price-spread quartiles.

Accurately evaluating the emissions impact of real time pricing requires high frequency data on marginal emissions rates and demand. The variation in the demand reduction implies using average emissions rates that lead to incorrect estimates. The environmental impacts of real time pricing are also sensitive to fuel prices. Setting a socially efficient subsidy to encourage the adoption of real time pricing by policy makers requires accurate forecasts of future fuel costs. It may also be possible to peg such a subsidy to fuel cost spreads to ensure that the real time pricing adjusts with the environmental benefits of the program, although this pegging adds significant administrative overhead.

The real time pricing exercise represents an application of this methodology, but this approach will be effective for at least two types of policy evaluations. The results can be applied to any policy or market shock that changes input prices. For example, a carbon tax could shift the relative price of generation from oil and natural gas generation in a manner similar to the natural gas price shock. The results are also useful for evaluating any

³⁶ The emissions of CO₂ are regulated under the Regional Greenhouse Gas Initiative (RGGI) a cap and trade program for generators in the Northeastern United States. Because the aggregate emissions are below the cap, these emission reductions can be considered a nonpecuniary benefit. If the cap is binding, then these reductions are private benefits to the electricity generator owners valued at the price of allowances.

energy or environmental policy that has heterogeneous impacts throughout the day. For example enhanced building codes, energy efficiency regulation and peak shaving programs all impact generation differently across hours within the day and therefore predicting the environmental impacts of those policies requires estimates of hourly marginal emissions and relative fuel prices.

7 Conclusion

In this paper, we describe a methodology for determining what fuel generators are using based on hourly data. We use hourly data on generator output, fuel inputs, and pollution outputs to estimate hour-by-hour which fuel type is in use. The results suggest that the fuel switching generators in NYC shifted to relying primarily on natural gas after 2008 when natural gas prices began to fall.

We then estimate how fuel prices have affected generation levels, fuel choices, and pollution emissions for NYC generators in aggregate and for two flexible fuel generators. The results suggest that while overall generation levels have not changed, flexible fuel generators switched heavily to natural gas at fuel price spreads of around \$4.40 per mmBTU. The flexible fuel generators dispatched somewhat less, used more natural gas, and polluted significantly less as natural gas became relatively cheaper. The environmental benefits of cheaper natural gas come primarily from fuel switching at flexible generators, which accounts for around two-thirds of their observed emission reductions. The lower levels of generation from these generators account for the remaining reductions in pollution.

These results are useful in evaluating the effect of fuel switching on both environmental and energy policies. The marginal emission profile for New York City has dropped across all demand levels. This drop implies that the environmental impacts of real-time electricity pricing, plug-in hybrid cars, or increased wind generation are likely different in the current high fuel-price spread environment. Current fuel price spreads are at, or beyond, the fourth quartile of the fuel prices we observe during our study period. Under current fuel prices, we are unlikely to observe any switch back to oil-fired generation; but under plausible scenarios, the price spread could drop into the first quartile of observed spreads (around \$4.40 per mmBTU). The EIA's Annual Energy Outlook for 2013 (EIA

(2013)) reports fuel prices, among many other variables, for 28 different future scenarios. Under the scenarios that include high productivity from new oil extraction techniques (including fracking and related technologies), the forecasted price spread drops to less than \$6.00.³⁷ The large scale export of liquid natural gas could lead to convergence between US and world prices in the natural gas market, which might lead to significant increases in gas prices and could return spreads to the first quartile. In these cases the environmental benefits of fuel switching could be lost.

New York City is an important electricity market with high levels of pollution emissions in a densely populated region. When considering the external validity of results, it is important to draw a distinction between the extensive and intensive margins. Intensive margin fuel switching, existing plants changing their fuel inputs, is limited by regions' significant oil-fired generating capacity. Most U.S. oil-fired generation is concentrated in the Northeast and many large cities in the region have reliability requirements. The extensive margin, changes in dispatch intensity of natural gas fired power plants, can and do occur anywhere in response to fuel price changes. The results presented here should be useful in evaluating the environmental impact of fuel switching in metropolitan areas with a mix of oil and natural gas-fired generation. The extensive margin results, which account for around a third of the environmental benefits of fuel switching, should prove helpful for general evaluations of the environmental benefits of fuel switching.

Recent volatility in oil prices and natural gas price spikes driven by pipeline capacity constraints could potentially reverse the results estimated here. Fuel switching at the extensive margin is particularly tenuous, but if oil prices remain low for a long enough period even the observed intensive margin fuel switching could be reversed. Evaluating the impact of environmental and energy policy requires dynamic modeling that considers fuel prices.

There are several recent papers that examine the impact of environmental regulation on a generator's fuel choice, but these results suggest that market driven changes in fuel prices can have significantly larger

³⁷ The EIA's "low oil price" scenario projects the price of fuel to fall to \$13.21 by 2040 while natural gas prices increase to \$7.23.

impacts. Changing fuel prices can have significant impacts on the environmental impact of existing energy policy. Our results suggest that going forward the literature using estimates of marginal emissions rate to evaluate environmental or energy policy should explicitly address the impacts of fuel prices.

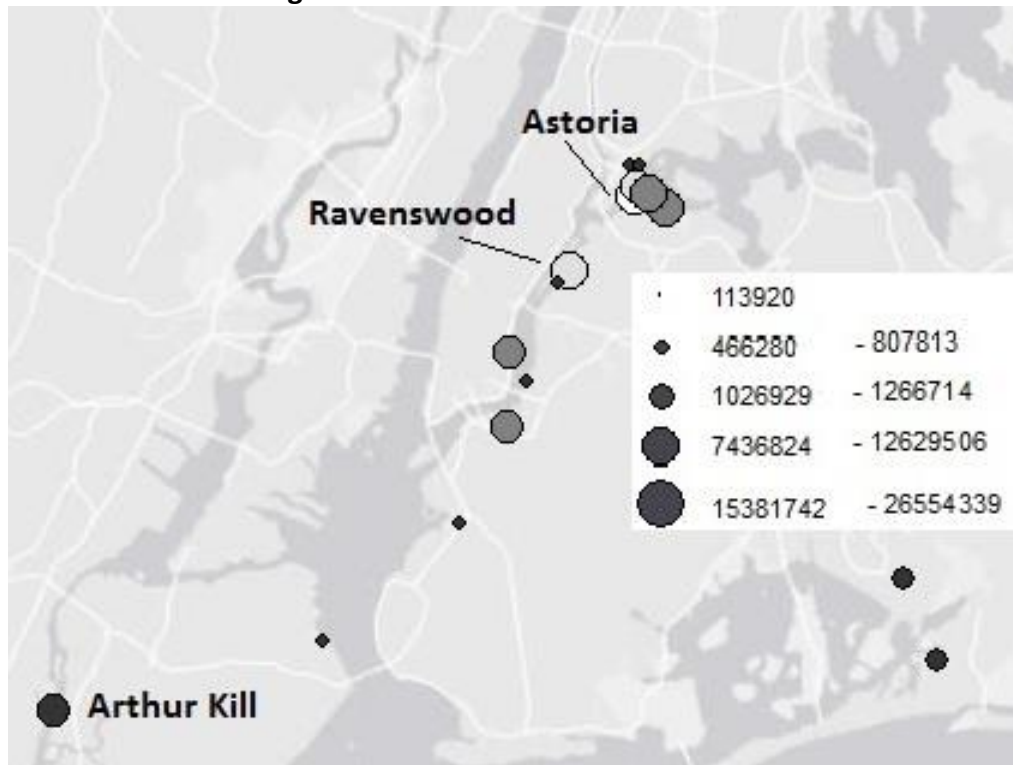
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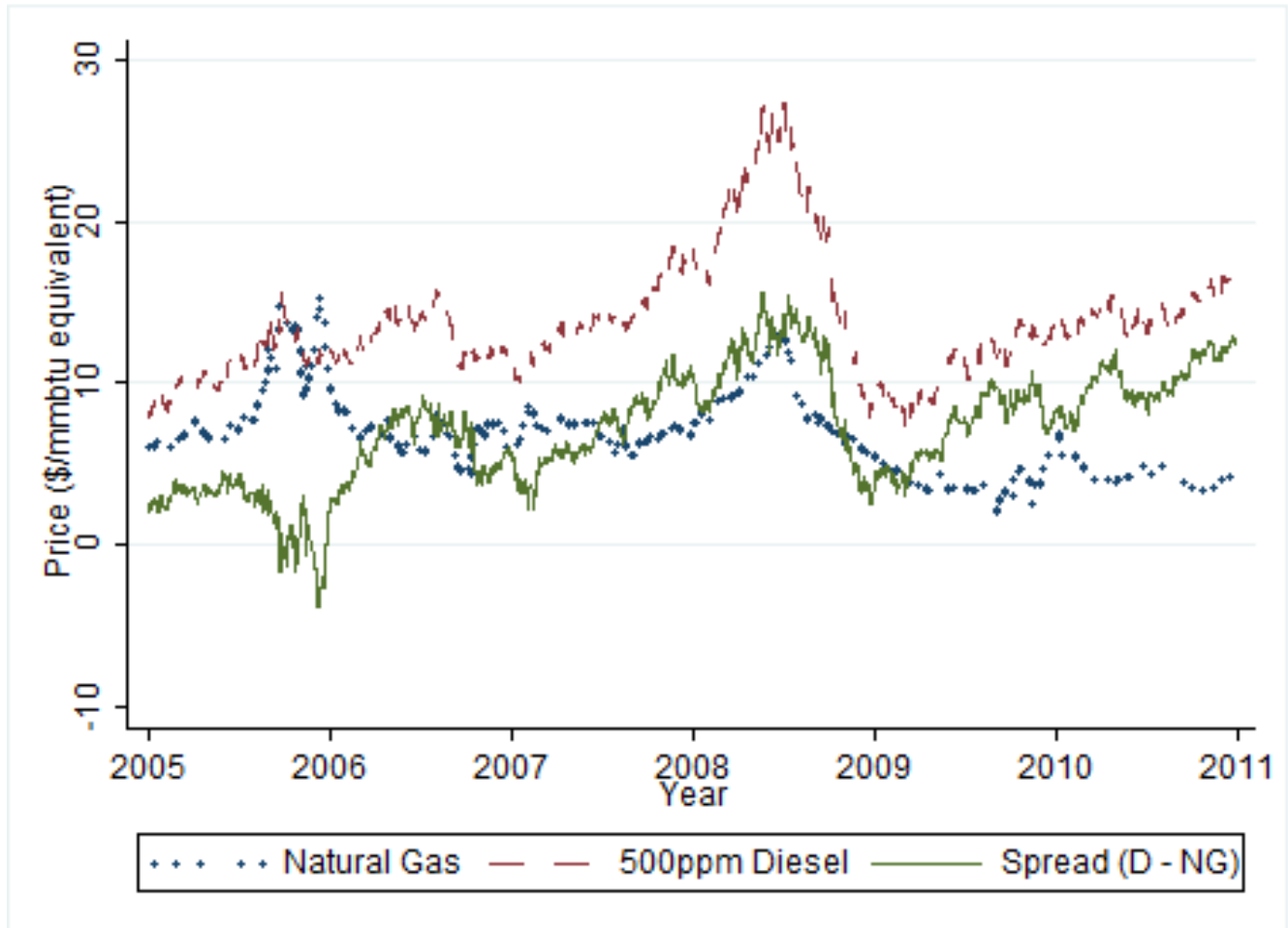
8 Figures and Tables

Figure 1: Generators in the NYC area



Note: New York City Electricity generator locations. The size of the circle represents the total generation at the generator from 2005-2010. The darkness of the circle represents the fraction of generation fueled by natural gas. White circles produce less than 50 % of the total generation from natural gas during the study period. Black circles produce all of their power from natural gas. Electricity generator locations provided by the EPA Clean Air Monitoring database. Total generation and fraction of natural gas are calculated by the authors. Background map provided by ERSI ArcMap.

Figure 2: Price of natural gas and oil over time



Note: Monthly price of natural gas at Henry Hub and diesel oil delivered in New York City both priced in \$/mmbtu equivalent. The solid line represents the spread between diesel and natural gas prices. Source: U.S. Energy Information Agency.

Table 1: Variable list and summary statistics

Variable	Mean	Std. Dev	Units
Generation	232.7	244.1	MWh
SO ₂	97.1	321.0	Pounds
NO _x	159.4	299.5	Pounds
CO ₂	146.2	144.3	Pounds
Heat Input	2337.2	2186.8	mmBTU

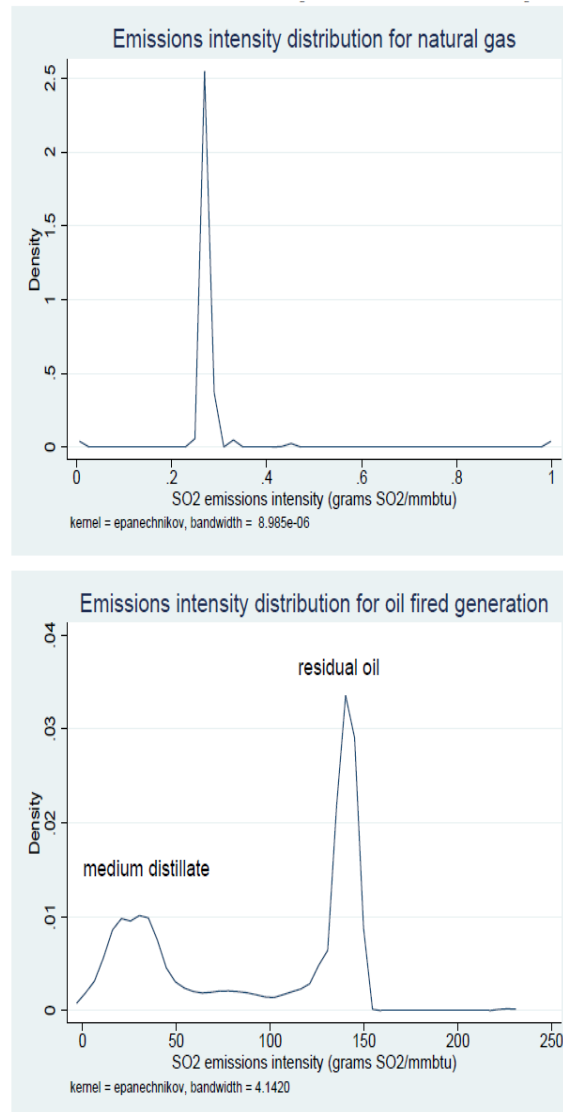
Note: Summary statistics across 528,765 electricity generator-hour observations. Generation is measured in megawatt-hours, pollution emissions are measured in pounds, and heat input is measured in millions of British thermal units (mmBTU), which describes the energy content of the fuel used in generation. Source: Authors' calculations from EPA Clean Air Markets data.

Table 2: Emissions rates of fuels

	NATURAL GAS	FUEL OIL
CO ₂	1135	1672
SO ₂	0.1	12
NO _x	1.7	4

Note: Pollution emissions measured in pounds per megawatt-hour of electricity generated. Source: US EPA EGrid 2000.

Figure 3: Emissions intensity distributions by fuel type



Note: The top panel displays the SO₂ emission intensity for the stacks in the range of 0 to 1 grams of SO₂/mmbTU. The bottom panel is the distribution in the oil-fired range of greater than 1 gram of SO₂/mmbtu. Each panel graphs the frequency against the emission rate measured in grams of SO₂/mmbTU.

Table 3: NYC Power Plant Generation by Fuel Type, 2005-2010

Facility	Gas (1)	Clean Dist. (2)	Medium Dist. (3)	Residual (4)	Fraction NYC gen (5)	Gen. (GWh) (6)
23rd and 3rd	100%	0%	0%	0%	1%	808
Arthur Kill	100%	0%	0%	0%	6%	7437
Astoria Energy	99%	1%	0%	0%	13%	16281
Astoria Gas Turbine Power	90%	0%	0%	10%	0%	114
Astoria Generating Station	39%	18%	29%	15%	15%	18571
Bayswater Peaking Facility	76%	24%	0%	0%	0%	513
Brooklyn Navy Yard Cogen	98%	2%	0%	0%	8%	9512
Charles Poletti	59%	7%	22%	11%	8%	10270
East River	100%	0%	0%	0%	10%	12630
Far Rockaway	100%	0%	0%	0%	1%	1267
Harlem River Yard	100%	0%	0%	0%	0%	466
Hell Gate	100%	0%	0%	0%	0%	522
KLAC Cogeneration	100%	0%	0%	0%	1%	1027
North 1st	100%	0%	0%	0%	0%	489
Poletti II (500 MW CC)	99%	1%	0%	0%	12%	15382
Pouch Terminal	100%	0%	0%	0%	0%	591
Ravenswood Generating Station	61%	2%	19%	17%	22%	26554
Vernon Boulevard	100%	0%	0%	0%	1%	626
NYC Total	78%	4%	10%	7%	100%	123060

Note: Generation by fuel type from 2005-2010. Column 1 is the fraction of generation from natural gas, column 2 from clean distillate, column 3 from medium distillate, column 4 from residual oil, and column 5 from the NYC load served by that plant, Column 6 is the total generation measured in gigawatt hours.

Table 4: Spread quartile summary

Quartile	Mean	Min	Max	SD
1	2.69	-3.74	4.39	1.63
2	5.97	4.40	7.54	0.93
3	8.57	7.54	9.68	0.61
4	11.66	9.68	15.53	1.36

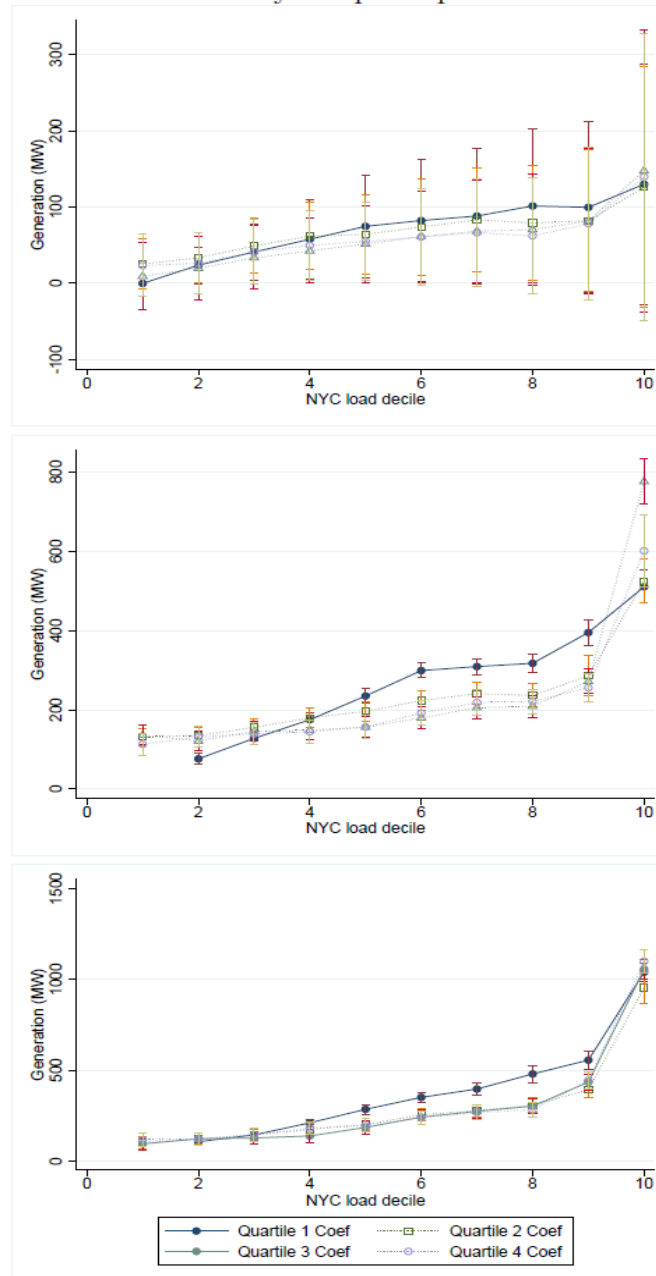
Note: The descriptive statistics of the fuel price spread, defined as (*diesel fuel - natural gas*) by fuel price spread quartile.

Table 5: Distribution of spread quartiles across time

Year	1	2	3	4	Total
2005	360	5	0	0	365
2006	71	192	102	0	365
2007	40	157	101	67	365
2008	31	39	56	240	366
2009	50	132	150	33	365
2010	0	19	140	206	365
Total	552	544	549	546	2,191

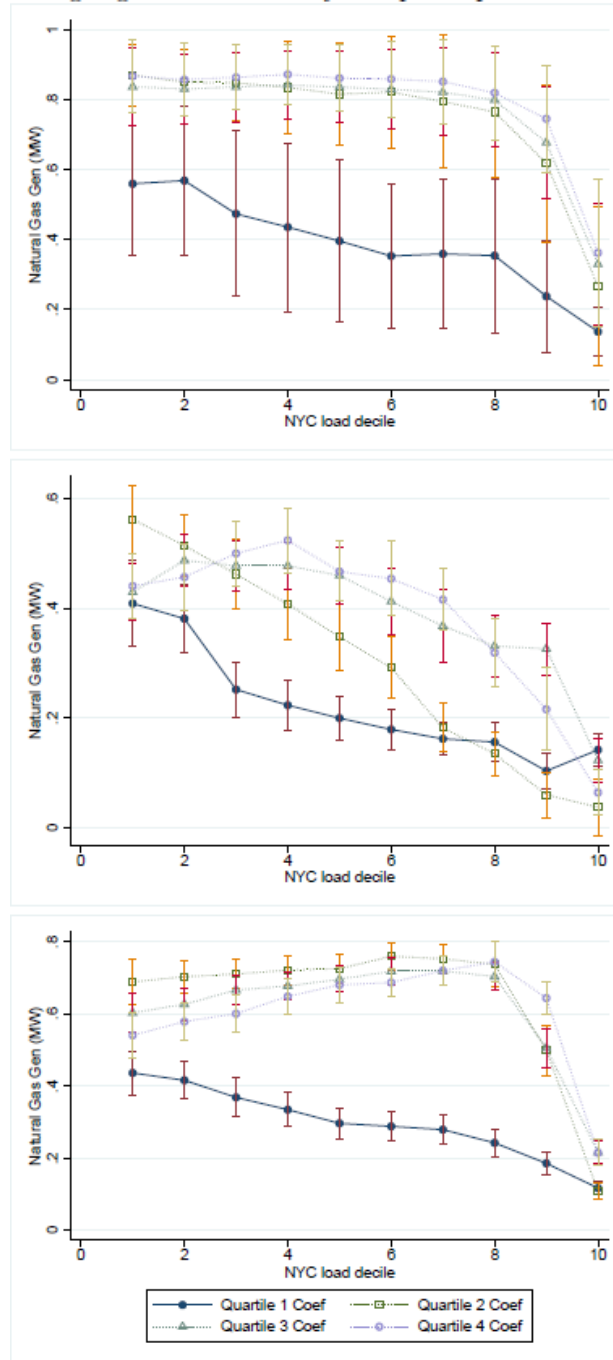
Note: The number of days for a year in each spread quartile. Spread quartile 1 occurs mostly in 2005, but the spread drops that low as late as 2009.

Figure 4: Generation level by fuel price spread and NYC load level



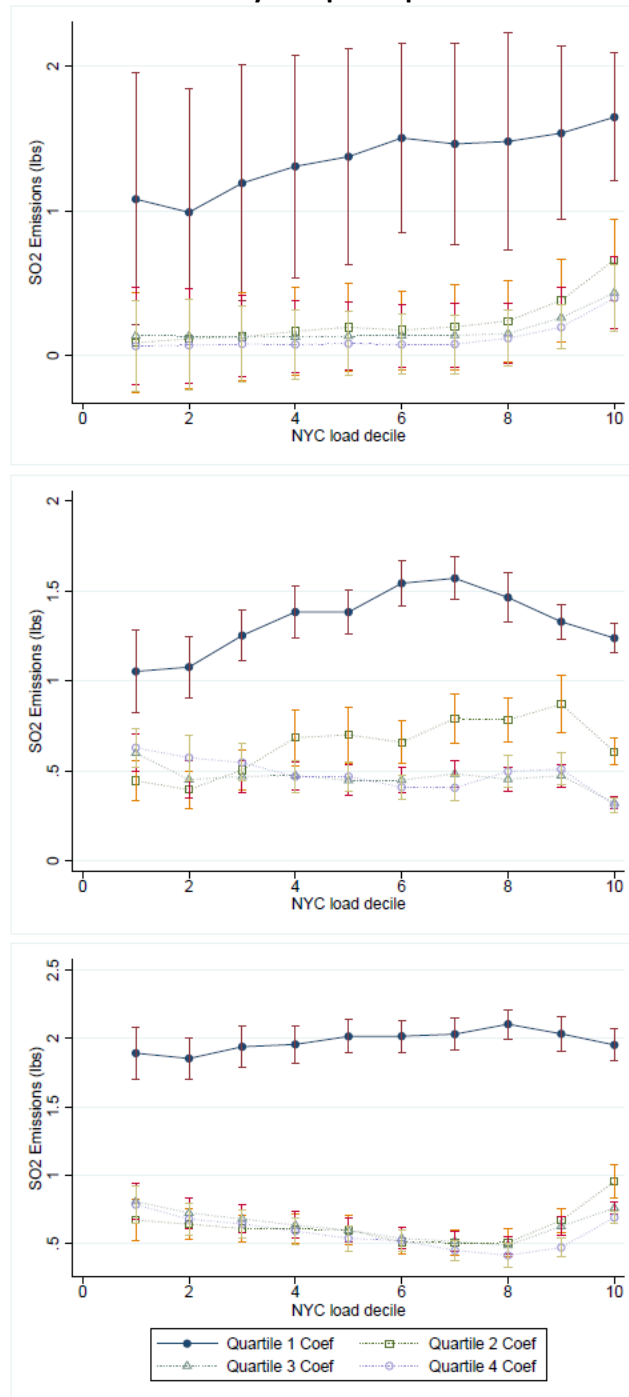
Note: The results of the regressions based on equation 1 that estimate the average NYC generation level conditional on the weather, month, and year fixed effects for all of the NYC generators (top panel), the Astoria Generating Station (middle panel), and the Ravenswood Generating Station (bottom panel). The vertical axis is the level of generation measured in megawatts. The vertical bars represent the 95% confidence interval for the coefficient estimate. All of the regressions include robust standard errors; and, in the regressions for all of the generators in NYC, the standard errors are clustered at the facility level. This clustering corrects for both intra-power plant correlation across time and possible serial correlation in the error term. The average generation level across the entire data set is 260.8 megawatts.

Figure 5: Marginal natural gas generation level by fuel price spread and NYC load level



Note: The results of the regressions that estimate the marginal contribution of natural gas-fired generation from a one megawatt increase in generation conditional on the weather, month, and the year fixed effects for all NYC generators (top panel) the Astoria Generating Station (middle panel), and the Ravenswood Generating Station (bottom panel). The vertical axis is the level of generation measured in megawatt-hours. The vertical bars represent the 95 % confidence interval for the coefficient estimate. All of the regressions include robust standard errors; and, in the regressions for all of the generators in NYC, the standard errors are clustered at the generator level. This error structure corrects for both the intra-electricity generator correlation across time and the possible serial correlation in the error term. The average natural gas generation level across the entire data set is 205.9 megawatt-hours.

Figure 6: SO₂ emissions by fuel price spread and NYC load level



Note: The results of the regressions that estimate the marginal SO₂ emissions from a one megawatt-hour increase in generation conditional on the weather, month, and the year fixed effects for all of the NYC generators (top panel), the Astoria Generating Station (middle panel), and the Ravenswood Generating Station (bottom panel). The vertical axis is the level of SO₂ measured in pounds. The vertical bars represent the 95% confidence interval for the coefficient estimate. All of the regressions include robust standard errors; and, in the regressions for all of the generators in NYC, the standard errors are clustered at the generator level. This error structure corrects for both the intra-electricity generator correlation across time and the possible serial correlation in the error term. The average hourly SO₂ emissions level across the entire data set is 87.2 pounds.

Table 6: Emissions and generation by quartile

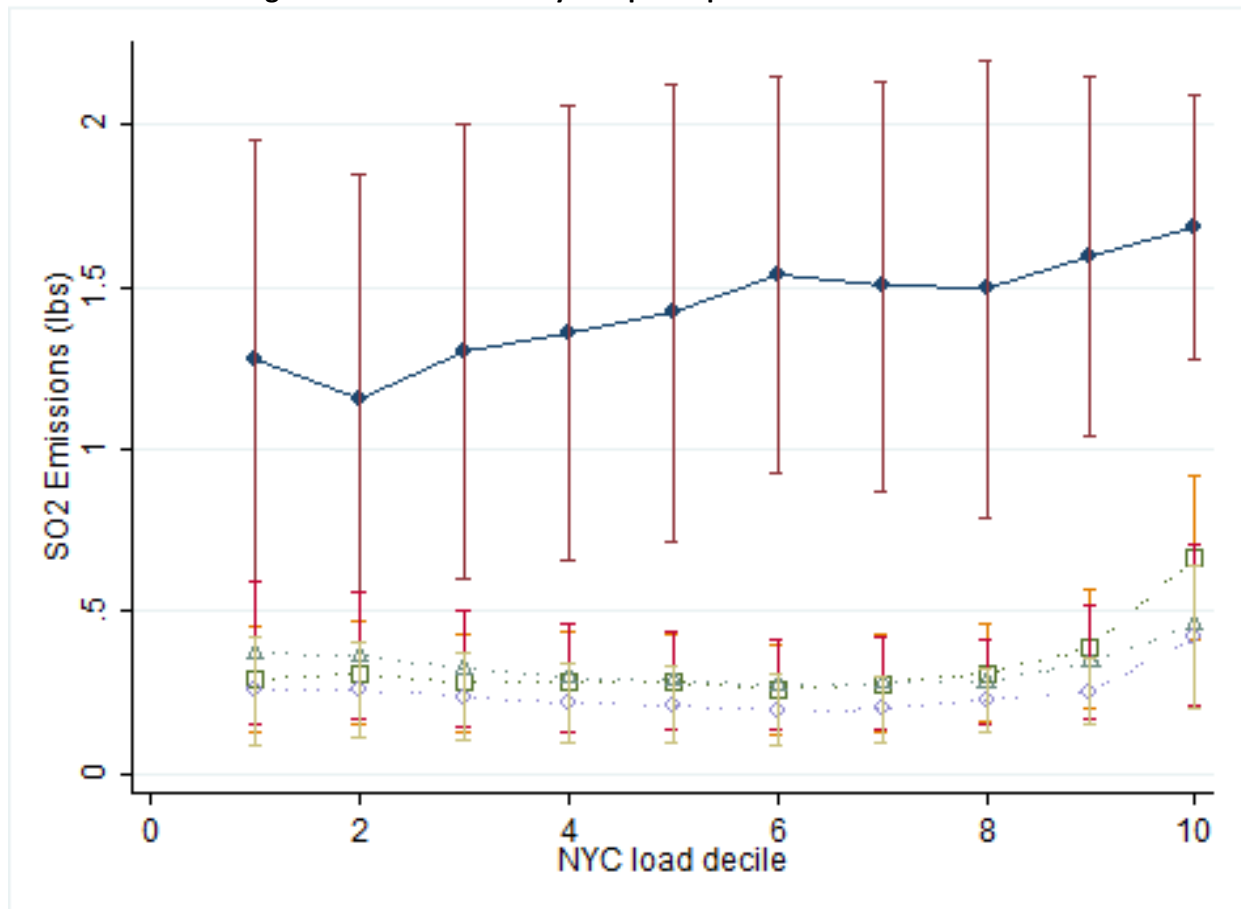
Quartile	SO2 Emissions Q1	SO2 Emissions Q2-4	SO2 Emissions Change	Gen Q1	Gen Q2-4	Extensive Margin	Intensive Margin
23rd and 3rd	0.5	0.6	0.2	160.2	215.6	0.2	0.0
Arthur Kill	6.0	5.8	-0.2	1,902.3	1,820.1	-0.3	0.0
Astoria Energy	7.2	9.6	2.4	2,096.2	4,726.9	9.0	-6.6
Astoria Gas Turbine Power	14.6	7.1	-7.6	9.3	34.9	40.1	-47.6
Astoria Generating Station	5,400.0	621.1	-4,778.9	6,027.3	4,087.3	-1,738.1	-3,040.8
Bayswater Peaking Facility	8.8	4.3	-4.4	140.3	118.9	-1.3	-3.1
Brooklyn Navy Yard Cogeneration	31.4	11.2	-20.2	2,604.1	2,253.3	-4.2	-16.0
Charles Poletti	1,865.3	286.8	-1,578.4	3,530.9	2,168.1	-720.0	-858.5
East River	419.7	148.6	-271.1	3,319.2	3,091.2	-28.8	-242.3
Far Rockaway	1.0	1.0	0.0	306.9	319.9	0.0	0.0
Harlem River Yard	0.4	0.3	-0.1	145.2	103.6	-0.1	0.0
Hell Gate	0.5	0.3	-0.1	157.2	117.5	-0.1	0.0
KIAC Cogeneration	0.3	0.9	0.6	120.8	302.0	0.5	0.0
North 1st	0.4	0.3	-0.1	138.2	116.6	-0.1	0.0
Poletti 500 MW CC	7.1	11.3	4.3	1,933.2	4,487.1	9.3	-5.1
Pouch Terminal	0.3	0.4	0.2	102.4	161.2	0.2	0.0
Ravenswood Generating Station	7,000.0	757.0	-6,243.0	8,566.8	5,893.1	-2,184.6	-4,058.3
Vernon Boulevard	0.5	0.4	-0.1	182.5	147.1	-0.1	0.0
Total	14,763.8	1,867.2	-12,896.6	31,443.0	30,164.4	-4,618.4	-8,278.2

Note: The levels of SO₂ emissions (in tons) and generation (in gigawatts) between quartile 1 and the average of quartiles 2-4. The extensive margin is the change in emissions due to changing generation levels. It is calculated from the change in generation evaluated at the quartile 1 emission rate. The intensive margin is the change in emissions from quartile 1 to quartiles 2-4 due to the changing emissions rate. It is calculated from the change in the emission rate calculated at the average quartile 2-4 generation level.

Table 7: Pollution permit prices

Spread Quartile	SO ₂	NO _x
1	750.31	1862.74
2	434.24	909.52
3	293.25	557.54
4	195.44	513.38

Note: Each column reports the average price for a specific type of pollution permit across the fuel-price spread quartiles. The column labeled SO₂ is the price of a permit to emit a ton of SO₂ in the Acid Rain Market. The column labeled NO_x is the price of a permit to emit a ton of NO_x during the ozone season.

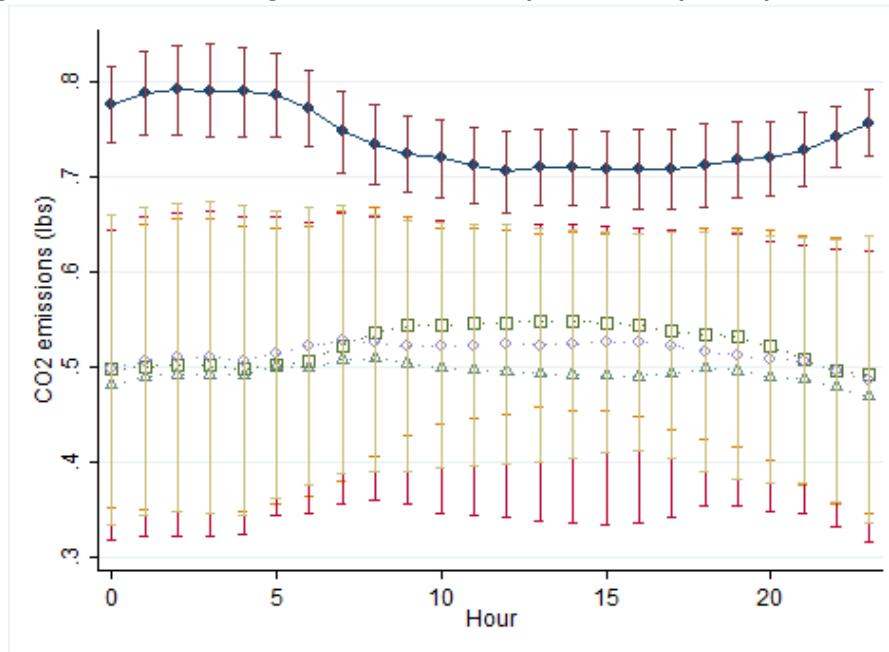
Figure 7: SO₂ emissions by fuel price spread and NYC load level

Note: The results of the regressions that estimate the marginal SO₂ emissions from a one megawatt increase in generation conditional on the weather, month, and the year fixed effects for all of the NYC generators. The regression includes the robust standard errors, and the standard errors are clustered at the generator level. This specification includes controls for both the SO₂ and NO_x permit prices

Table 8: Demand impacts of real time pricing by Hour of Day and Fuel-Price Spread Quartile

	No Summer				Summer			
	Spread Quartile							
Hour	1	2	3	4	1	2	3	4
6	-591	-476	-461	-516	-510	-388	-369	-532
7	-637	-517	-497	-554	-568	-452	-416	-623
8	-678	-553	-528	-597	-628	-506	-458	-705
9	-723	-586	-561	-636	-684	-547	-493	-750
10	-733	-595	-571	-644	-706	-574	-523	-792
11	-739	-594	-576	-647	-752	-602	-551	-829
12	-735	-588	-574	-637	-795	-630	-583	-860
13	-730	-584	-575	-636	-847	-679	-626	-905
14	-726	-572	-570	-631	-864	-712	-660	-956
15	-726	-572	-576	-634	-882	-738	-684	-1005
16	-769	-601	-609	-671	-890	-753	-691	-1033
17	-852	-633	-648	-709	-850	-720	-653	-987
18	-829	-635	-640	-697	-769	-646	-592	-881
19	-786	-635	-628	-698	-714	-603	-551	-833
20	-737	-617	-593	-672	-703	-597	-543	-839
21	-677	-557	-537	-598	-693	-578	-525	-809

Note: The estimated reductions in demand under the real time pricing of electricity in New York City across the hour, price-spread quartile, and season. The demand reductions are measured in megawatts. The average hourly demand over the full sample is 9,166 megawatts.

Figure 8: Estimated Marginal CO₂ Emissions by Hour of Day and Spread Quartile

Note: The results of the regressions that estimate the marginal CO₂ emissions from a one megawatt increase in generation conditional on the weather, month, and the year fixed effects for all of the NYC generators for each hour of the day and price-spread quartile. The regression includes the robust standard errors, and the standard errors are clustered at the generator level.

Table 9: Estimated CO₂ Emissions Reductions Under Real Time Pricing

Hour	No Summer				Summer			
	Spread Quartile							
	1	2	3	4	1	2	3	4
6	-365	-276	-295	-278	-465	-238	-231	-265
7	-398	-298	-317	-302	-492	-262	-248	-290
8	-424	-320	-338	-321	-507	-289	-268	-316
9	-454	-343	-362	-339	-531	-314	-285	-334
10	-462	-351	-369	-341	-531	-323	-288	-336
11	-466	-354	-370	-342	-532	-323	-288	-338
12	-461	-353	-368	-341	-524	-321	-285	-333
13	-457	-350	-367	-337	-515	-319	-285	-333
14	-453	-348	-365	-336	-516	-314	-281	-330
15	-454	-348	-364	-335	-516	-313	-283	-332
16	-482	-369	-385	-356	-545	-328	-299	-353
17	-535	-407	-425	-393	-604	-344	-318	-373
18	-521	-395	-412	-381	-588	-341	-316	-364
19	-490	-373	-390	-360	-560	-339	-313	-360
20	-457	-349	-363	-336	-530	-328	-295	-344
21	-418	-318	-331	-309	-488	-291	-263	-304

Note: The estimated reductions in CO₂ (measured in tons) emissions under real time pricing of electricity in New York City across hour, price-spread quartile, and season. The estimates are generated by multiplying the marginal emissions in an hour and fuel-price spread quartile by demand reduction in the same hour and price-spread quartile. The average hourly CO₂ emissions across the full sample are 1,383 pounds.